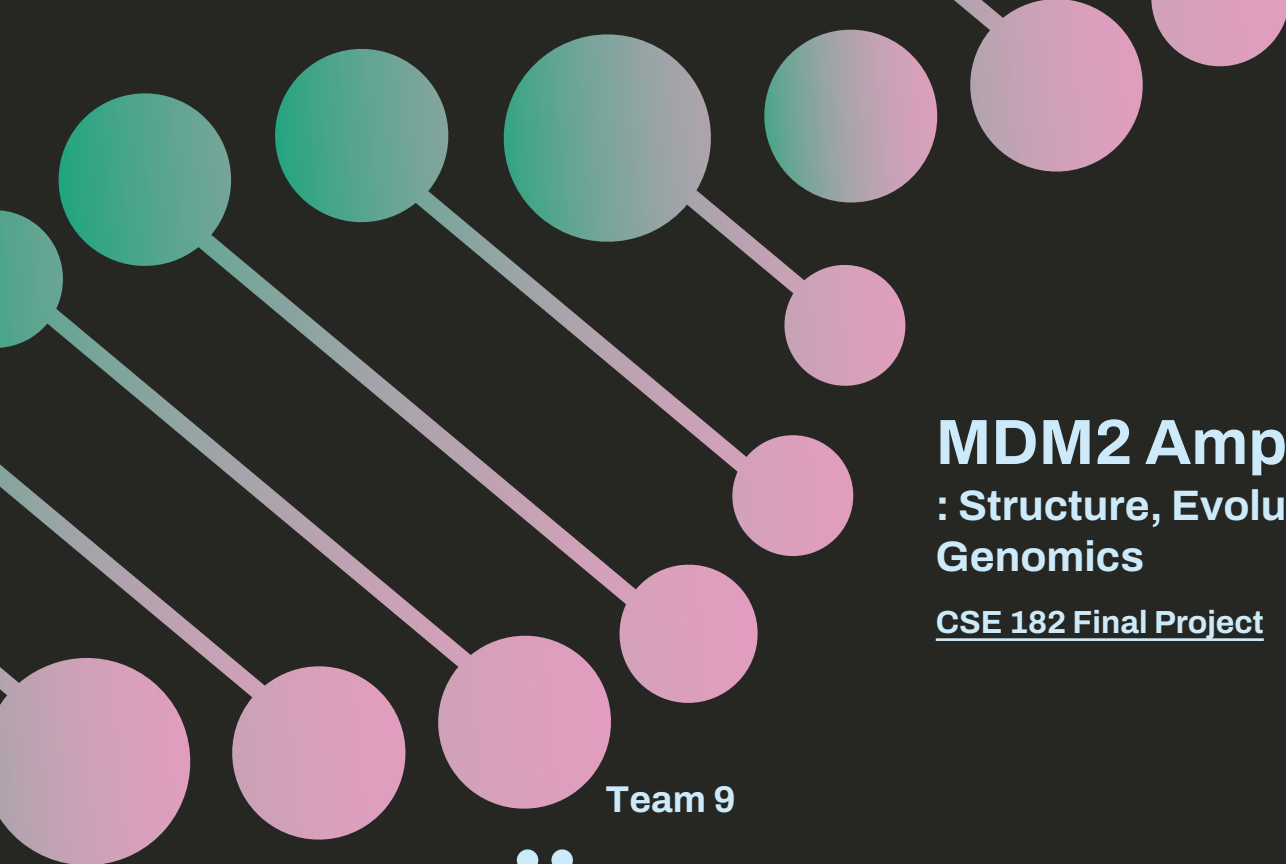




MDM2 Amplification in ecDNA

: Structure, Evolution, and Impact on Tumor Genomics

CSE 182 Final Project

Team 9



Evelyn Quan, James Caviness, Timothy Hsiao, Leena Khattat, Lydia Roh

01.

Gene Details

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MDM2

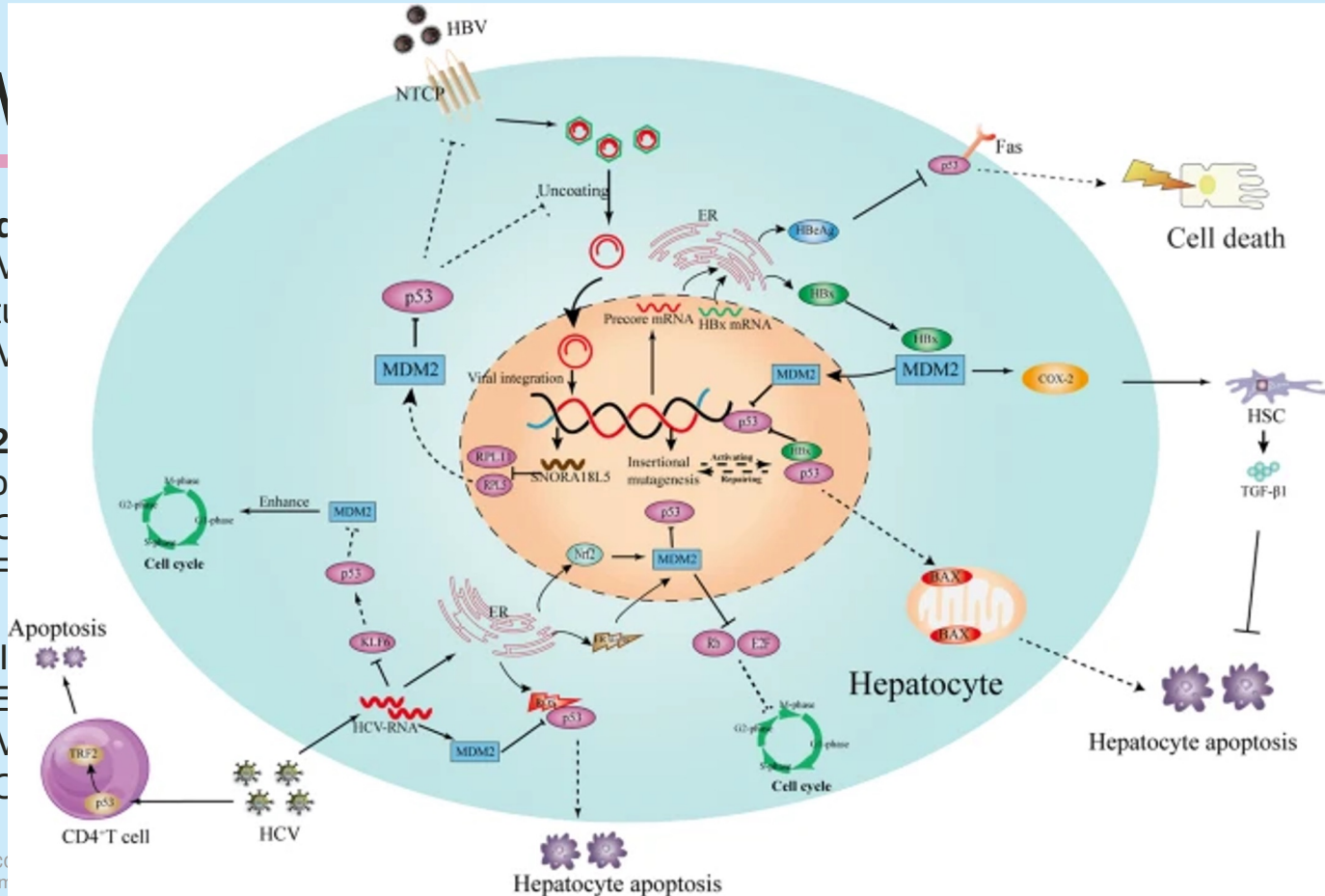
Onco

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ERα I

- E
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- C

Wu L, Maki C
Available from



p53

a

Hepatocyte

Hepatocyte apoptosis

Hepatocyte apoptosis

How Ancient is MDM2?

- Arose from a gene duplication event prior to the emergence of bony vertebrates more than **440 million** years ago

Most distant **Orthologs**

- Atlantic hagfish (*Myxine glutinosa*) found through protein-protein BLAST search; approx. **310 M** years
- The invertebrate MDM homologs are in Sea squirts (*C. intestinalis*); over **500 M** years



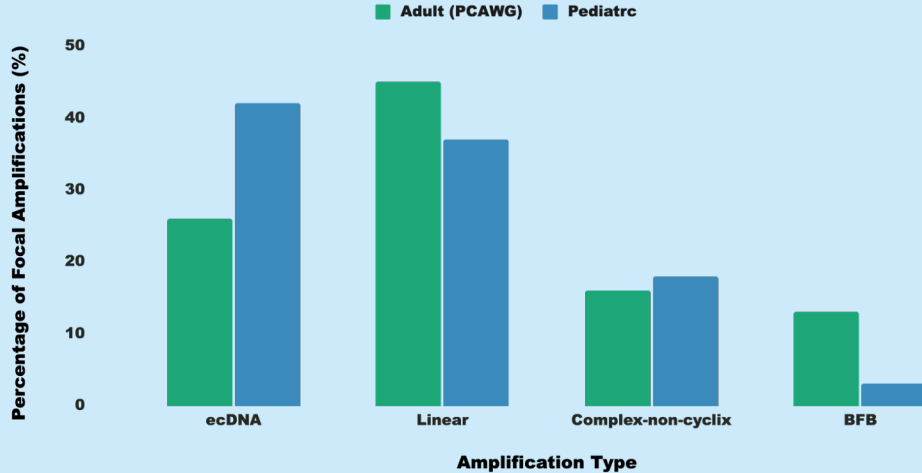
• Leucoraja erinaceus	sharks & rays	636	4	Leucoraja erinaceus hits
• Stegostoma tigrinum	sharks & rays	636	9	Stegostoma tigrinum hits
• Amblyraja radiata	sharks & rays	623	5	Amblyraja radiata hits
• Callorhinchus milii	chimaeras	607	5	Callorhinchus milii hits
• Myxine glutinosa	hagfishes	421	2	Myxine glutinosa hits



MDM2 Amplification Types

(Adult vs Pediatric Tumors)

Percentage of MDM2 Amplification Types in Pediatric vs Adult Tumors

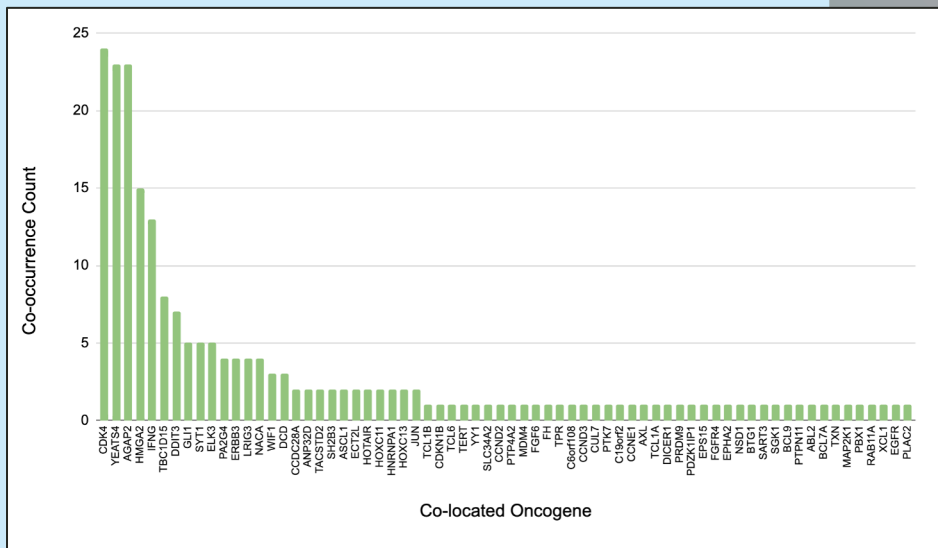


Adult (n = 2703)
Pediatric (n = 1094)

MDM2 Co-Occurring Genes in ecDNA (PCAWG)

Gene	Count	Fraction	Distance to MDM2 (bp)	
CDK4	24	0.83	11076842	
YEATS4	23	0.79	548446	
AGAP2	23	0.79	11094891	
HMGA2	15	0.52	2931473	
IFNG	13	0.45	669606	
TBC1D15	8	0.28	3055176	
DDIT3	7	0.24	11308307	
GLI1	5	0.17	11360834	
SYT1	5	0.17	10331531	
ELK3	5	0.17	27405239	
PA2G4	4	0.14	12717626	
ERBB3	4	0.14	12735025	
LRIG3	4	0.14	9930531	
NACA	4	0.14	12107997	
WIF1	3	0.1	3740895	
DCD	3	0.1	14180383	
Rare MDM2 co-occurrences (≤2 samples)	52 genes	0.03~0.07	∞	37
			Ch12	14

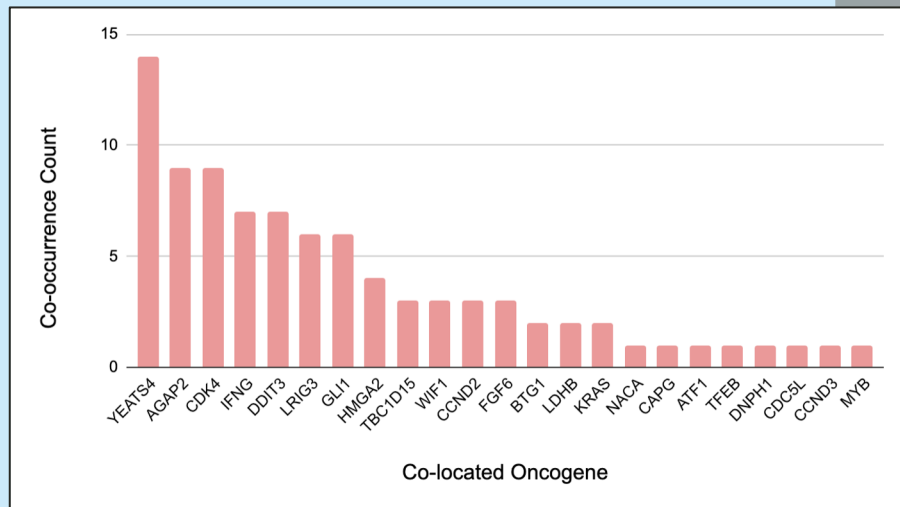
Total 29 ecDNA amplicons
Total of 68 co-occurrence genes

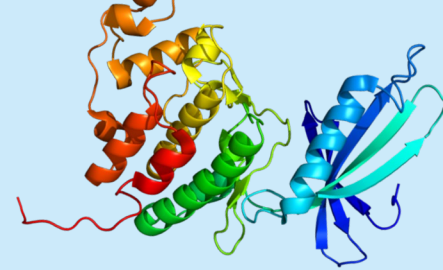


MDM2 Co-Occurring Genes in ecDNA (Pediatric)

Gene	Count	Fraction	Distance to MDM2 (bp)
YEATS4	14	0.93	548446
AGAP2	9	0.6	11094891
CDK4	9	0.6	11076842
IFNG	7	0.47	669606
DDIT3	7	0.47	11308307
LRIG3	6	0.4	9930531
GLI1	6	0.4	11360834
HMGA2	4	0.27	2931473
TBC1D15	3	0.2	3055176
WIF1	3	0.2	3740895
CCND2	3	0.2	64602824
FGF6	3	0.2	64362362
BTG1	2	0.13	23316202
LDHB	2	0.13	47180268
KRAS	2	0.13	43598773
Rare MDM2 co-occurrences (=1 samples)	8 genes	0.07	∞ 6 Ch12 2

Total 15 ecDNA amplicons
Total of 23 co-occurrence genes





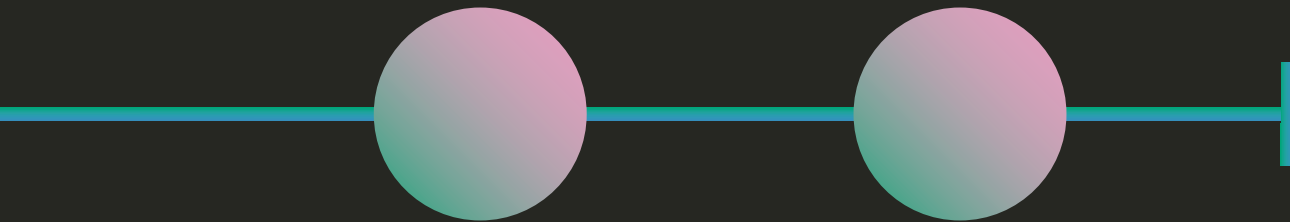
Adult (PCAWG)

Gene	Count	Fraction	Distance to MDM2 (bp)	
CDK4	24	0.83	11076842	
YEATS4	23	0.79	548446	
AGAP2	23	0.79	11094891	
HMGA2	15	0.52	2931473	
IFNG	13	0.45	669606	
TBC1D15	8	0.28	3055176	
DDIT3	7	0.24	11308307	
GLI1	5	0.17	11360834	
SYT1	5	0.17	10331531	
ELK3	5	0.17	27405239	
PA2G4	4	0.14	12717626	
ERBB3	4	0.14	12735025	
LRIG3	4	0.14	9930531	
NACA	4	0.14	12107997	
WIF1	3	0.1	3740895	
DCD	3	0.1	14180383	
Rare MDM2 co-occurrences (≤2 samples)	52 genes	0.03~0.07	∞	37
			Ch12	14

Pediatric

Gene	Count	Fraction	Distance to MDM2 (bp)	
YEATS4	14	0.93	548446	
AGAP2	9	0.6	11094891	
CDK4	9	0.6	11076842	
IFNG	7	0.47	669606	
DDIT3	7	0.47	11308307	
LRIG3	6	0.4	9930531	
GLI1	6	0.4	11360834	
HMGA2	4	0.27	2931473	
TBC1D15	3	0.2	3055176	
WIF1	3	0.2	3740895	
CCND2	3	0.2	64602824	
FGF6	3	0.2	64362362	
BTG1	2	0.13	23316202	
LDHB	2	0.13	47180268	
KRAS	2	0.13	43598773	
Rare MDM2 co-occurrences (=1 samples)	8 genes	0.07	∞	6
			Ch12	2

Part 2: Gene Structure



Protein Sequence

Homo sapiens MDM2 proto-oncogene (MDM2), transcript variant 1, mRNA
NCBI Reference Sequence: NM_002392.6

MVRSRQMCNTNMSVPTDGAVTTSQIPASEQETLVRPKPLLLKLLKSVGAKDYTMKEVLFFYLGQYIMTKRLYDE
KQQHIVYCSNDLLGDLFGVPSFSVKEHRKIYTMIRNLVVVNQQESSDSGTSVSENRCHEGGSDQKDLVQELQ
EEKPSSSHLVSRPSTSSRRRAISETTEENSDLSGERQRKRHKSDSISLSFDESLALCVIREICCERSSSSSESTGTP
SNPDLDAGVSEHSGDWLDQDSVSDQFSVEFEVESLDSYSLSEEGQELSDDEDDEVYQVTVYQAGESDTSF
EEDPEISLADYWKCTSCNEMNPPLPSHCNRCWALRENWLPEDKGKDKGEISEKAKLENSTQAEFGDVPDCKK
TIVNDSRESCVEENDDKITQASQSQESEDYSQPSTSSSIYSSQEDVKEFEREETQDKEESVESSLPLNAIEPCVI
CQGRPKNGCIVHGKTGHLMACFTCAKKLKKRNKPCPVCRRQPIQMIVLTYFP

Sequence Length: 497 Amino Acids

Genomic Region

Sample Name

DO10960-SP23739

Feature Count

3

Oncogenes

CHIC2, EGFR, MDM2, PDGFRA

Classifications

ecDNA

Show

10

entries

Search:

Amplicon	Feature ID	Classification	Oncogenes	Location	Downloads
1 	25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_ecDNA_1	ecDNA	CHIC2, PDGFRA	4:54373631-55293946	BED PDF PNG
2 	25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_ecDNA_1	ecDNA	EGFR	7:54720334-55286759, 7:55289060-55305509	BED PDF PNG
3 	25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_ecDNA_1	ecDNA	MDM2	12:69110024-69386333	BED PDF PNG

Showing 1 to 3 of 3 entries

Previous

1

Next

- Precise Position | Chromosome 12: 69110024–69386333 (276309 Nucleotides)

Genomic Structure (1)

- 10 hits found BLASTing MDM2 protein sequence against chromosome 12 range 69110024-69386333 (GRCh37/hg19):
 - 69202987 — 69203076
 - 69207371 — 69207412
 - 69210590 — 69210748
 - 69214104 — 69214154
 - 69218143 — 69218211
 - 69218336 — 69218431
 - 69222551 — 69222712
 - 69229610 — 69229768
 - 69230453 — 69230530
 - 69233043 — 69233627



Genomic Structure (2)

blast_Start	blast_End	ref_Start	ref_End	start_Diff	end_Diff
69202987	69203076	69202988	69203073	-1	3
69207371	69207412	69207334	69207409	37	3
69210590	69210748	69210592	69210726	-2	22
69214104	69214154	69214105	69214155	-1	-1
69218143	69218211	69218143	69218211	0	0
69218336	69218431	69218335	69218432	1	-1
69222551	69222712	69222551	69222712	0	0
69229610	69229768	69229609	69229765	1	3
69230453	69230530	69230452	69230530	1	0
69233043	69233627	69233054	69233924	-11	-5697

Score	Expect	Method	Identities	Positives	Gaps	Frame
102 bits(254)	1e-24	Compositional matrix adjust.	45/53(85%)	51/53(96%)	0/53(0%)	+3
Query	58	EVLFYLGQYIMTKRLYDEKQQHIVYCSNDLLGDLFGVPSFSVKEHRKIYTMIIY				110
		+VLFYLGQYIMTKRLYDEKQQHIVYCSNDLLGDLFGVPSFSVKEHR			+++++	
Sbjct	100566	QVLFYLGQYIMTKRLYDEKQQHIVYCSNDLLGDLFGVPSFSVKEHR*FFSLVH				100724

- Large UTR:

Human Gene MDM2 (ENST00000258149.11_7) from GENCODE V48lift37

Description: MDM2 proto-oncogene, transcript variant 1 (from RefSeq NM_002392.6)

Gencode Transcript: ENST00000258149.11_7

Gencode Gene: ENSG00000135679.27_17

Transcript (Including UTRs)

Position: hg19 chr12:69,201,957-69,239,324 **Size:** 37,368 **Total Exon Count:** 11 **Strand:** +

Coding Region

Position: hg19 chr12:69,202,258-69,233,629 **Size:** 31,372 **Coding Exon Count:** 11



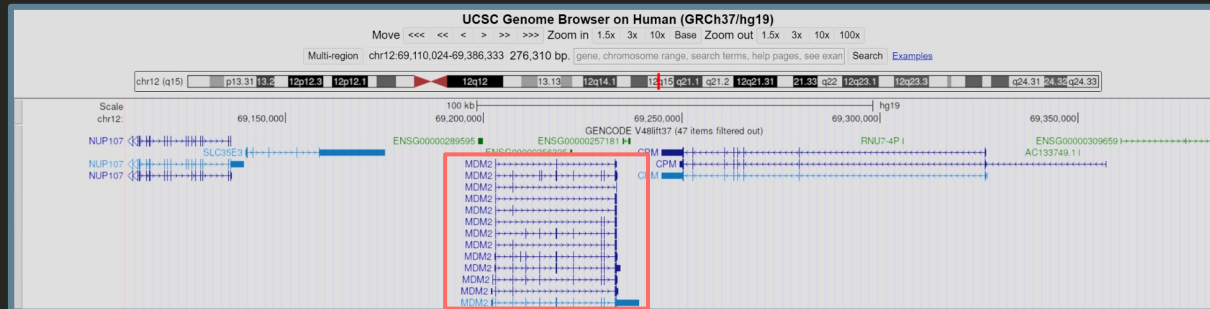
Genomic Structure (3)

- Is the gene truncated in ecDNA? Not Necessarily.
- Missing BLAST exon:

hg19 ENST00000258149.11_7

```
>hg19_knownGene_ENST00000258149.11_7_0 range=chr12:69201957-69202257 5'pad=0 3'pad=0 strand=+ repeatMasking=none
GGGCGCGCACCGAGGCACGCGGCGAGCTTGGCTGCTTCTGGGGCCCTGTG
TGGCCCTGTGTGTCTCGGAAAGATGGAGCAAGAAGCCGAGCCCGAGGGGCGG
CCGCGACCCCTCTGACCGAGATCCTGCTGCTTTTCGACGCCAGGAGCACCG
TCCCTCCCCGGATTAGTGCGTACGAGCGCCCAAGTGCCTTGCCCGGAGAG
TGGAATGATCCCCGAGGCCAGGGCGTCGTCTCCGCGCGCCCGTGAA
GGAACCTGGGGAGTCTTGAGGGACCCCGACTCCAAGCGCGAAACCCCG
G
>hg19_knownGene_ENST00000258149.11_7_1 range=chr12:69202258-69202271 5'pad=0 3'pad=0 strand=+ repeatMasking=none
ATGGTGAGGAGCAGC
>hg19_knownGene_ENST00000258149.11_7_2 range=chr12:69202272-69202987 5'pad=0 3'pad=0 strand=+ repeatMasking=none
```

- ecDNA range:



BEST SELLERS



PLANET VENUS

Venus has a beautiful name and is the second planet from the Sun. It's terribly hot



PLANET MERCURY

Mercury is the closest planet to the Sun and the smallest one in the Solar System

OUR LOCATIONS



MARS

Mars is actually a cold place



VENUS

Venus has a beautiful name

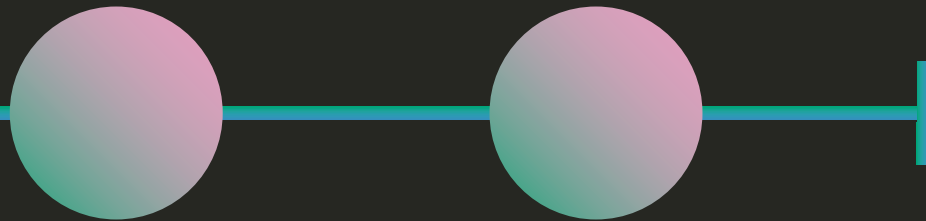


MERCURY

It's the closest planet to the Sun



Part 3: Amplicon Graph Analysis



Code and
output files



Amplicon Graphs

- DO10960-SP23739
- DO13474-SP28581
- DO219945-SP121763
- DO10970-SP23754
- DO14049-SP29697
- DO219949-SP121774
- DO11447-SP24708
- DO218742-SP117983
- DO13348-SP28335
- DO219944-SP121761

Identifying Structural Variants

- From the PCAWG data, we used Unix to extract all discordant edges from the relevant amplicon graph file in each dataset.

Extracting Discordant Edges

```
timti@Timpot:~$ cd Projects
timti@Timpot:~/Projects$ cd CSE182_Final
timti@Timpot:~/Projects/CSE182_Final$ ls
D010960-SP23739          D013348-SP28335          D0218742-SP117983       D0219949-SP121774
D010960-SP23739_amplicon3_discordant_edges.txt  D013348-SP28335_amplicon1_discordant_edges.txt  D0218742-SP117983_amplicon3_discordant_edges.txt  D0219949-SP121774_amplicon1_discordant_edges.txt
D010970-SP23754          D013474-SP28581          D0219944-SP121761       discordant_edges.txt:Zone.Identifier
D010970-SP23754_amplicon1_discordant_edges.txt  D013474-SP28581_amplicon2_discordant_edges.txt  D0219944-SP121761_amplicon1_discordant_edges.txt
D011447-SP24708          D014049-SP29697          D0219945-SP121762       discordant_edges.txt
D011447-SP24708_amplicon1_discordant_edges.txt  D014049-SP29697_amplicon1_discordant_edges.txt  D0219945-SP121763
timti@Timpot:~/Projects/CSE182_Final$ cd D010960-SP23739
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739$ ls
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_ecDNA_1  D010960-SP23739_CNV_CALLS.bed  D010960-SP23739_result_data.tsv  aa_directory
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_ecDNA_1  D010960-SP23739_classification_bed_files  D010960-SP23739_sample_metadata.json  aa_directory.tar.gz
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_ecDNA_1  D010960-SP23739_result_data.json  D010960-SP23739_sashimi_plots  cnvkit_directory.tar.gz
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739$ gunzip aa_directory.tar.gz
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739$ ls
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_ecDNA_1  D010960-SP23739_CNV_CALLS.bed  D010960-SP23739_result_data.tsv  aa_directory
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_ecDNA_1  D010960-SP23739_classification_bed_files  D010960-SP23739_sample_metadata.json  aa_directory.tar
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_ecDNA_1  D010960-SP23739_result_data.json  D010960-SP23739_sashimi_plots  cnvkit_directory.tar.gz
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739$ tar -xf aa_directory.tar
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739$ ls
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results  D010960-SP23739_CNV_CALLS.bed  D010960-SP23739_sample_metadata.json  cnvkit_directory.tar.gz
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25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_ecDNA_1  D010960-SP23739_result_data.json  aa_directory
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_ecDNA_1  D010960-SP23739_result_data.tsv  aa_directory.tar
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739$ cd 25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739/25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results$ ls
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_log  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_edges.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1.pdf  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_graph.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1.png  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_logs.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_cycles.txt  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3.pdf
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_edges.txt  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3.png
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_graph.txt  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_cycles.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon1_logs.txt  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_edges.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2.pdf  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_graph.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2.png  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon3_logs.txt
25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_amplicon2_cycles.txt  25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_summary.txt
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739/25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results$ cat 25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results.txt
3f6e8b7ab_amplicon3_graph.txt | grep "discordant" > D010960-SP23739_discordant_edges.txt
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739/25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results$ cat D010960-SP23739_discordant_edges.txt
discordant      12:693863324->12:69110024-      2_745773207566685      5      None      None
timti@Timpot:~/Projects/CSE182_Final/D010960-SP23739/25d3e2fc-7125-5426-b489-3e4b91814dfc__36548ada-345d-5258-ba55-ac43f6e8b7ab_AA_results$
```

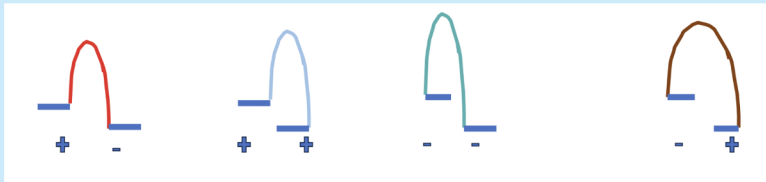
Clustering Discordant Edges

- We used a cutoff of 1000 base pairs to cluster edges into the file "clustered_edges.tsv", with the program, "cluster_edges.py" much like in our assignment for class
- This resulted in 1298 edges becoming 1248 clusters, indicating that we had many unique edges
- The file "clustered_edges.tsv" stores information about each edge: Chromosome, Location, and Direction of each endpoint, as well as frequency of that edge after clustering

clustered_edges.tsv > data							
1	chrom1	pos1	dir1	chrom2	pos2	dir2	count
12	12	59417889	-	12	58038280	-	1
13	12	60061068	+	12	58165054	-	1
14	12	60413626	-	12	60412348	+	2
15	12	60515452	+	12	59456518	+	1
16	12	60525035	-	12	60521844	+	3

Identifying Structural Variants

- Parsed "filtered_edges.tsv" to identify SVs from filtered discordant edges
- 32 inversions, 16 circularizations, 6 translocations, 17 deletions
- Used edge signs for identification:
 - Translocation = (+/-) and different start/end chromosomes
 - Deletion = (+/-) within same chromosome
 - Inversion
 - Right foldback (+/+)
 - Left foldback (-/-)
 - Circularization = (-/+)

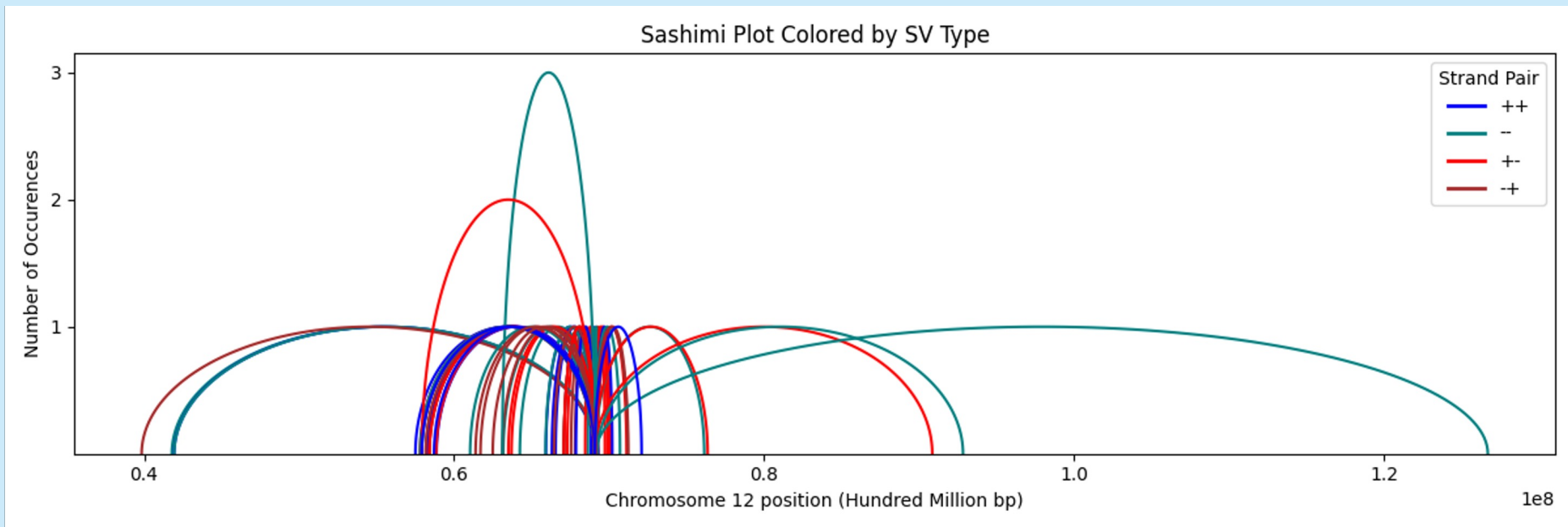


ID	Start_Position	End_Position	SV_Type	Frequency
1	12:69263448+	12:57498940+	Inversion	1
2	12:69728847+	12:69127970-	Deletion	1
3	12:69029013+	12:58292593+	Inversion	1
4	12:69115956+	12:68833902+	Inversion	1
5	12:69199255-	12:64240804-	Inversion	1
6	12:70087232+	12:69114474-	Deletion	1
7	12:69039623-	12:66546976+	Inversion	1
8	12:69039859-	6:48196328-	Translocation	1
9	12:69097355+	12:67381848-	Deletion	1
10	12:69175403-	12:67798314+	Circularization	1
11	12:69189407+	12:63499807-	Deletion	1
12	12:69248079+	6:48206489+	Translocation	1
13	12:69240420-	12:69187925-	Inversion	1
14	12:69124639-	12:48781024-	Inversion	1
15	12:69154302+	12:41933038+	Inversion	1
16	12:69247441-	12:41938811-	Inversion	1
17	12:69256516-	12:69255778+	Circularization	1
18	12:69278631-	12:39835182+	Circularization	1
19	12:69038267-	12:68630568-	Inversion	1
20	12:69089897-	12:58191819+	Circularization	1
21	12:69089897-	7:55711767-	Translocation	1
22	12:69151686-	12:58400705+	Circularization	1
23	12:69181977-	12:63171864-	Inversion	1
24	12:69189530-	7:55231664+	Translocation	1
25	12:69189530-	12:6945314+	Circularization	1
26	12:69251023+	7:55608304-	Translocation	1
27	12:69263935+	12:58863385+	Inversion	1
28	12:6924014+	12:68459805-	Deletion	1
29	1:162561637-	12:69217699+	Translocation	1
30	12:69028944+	12:57797526-	Deletion	1
31	12:69028954+	12:66386888+	Inversion	1
32	12:69040666+	12:68521989+	Inversion	1
33	12:69042649-	12:57797616-	Inversion	1
34	12:69043626-	12:65898283-	Inversion	1
35	12:69225027-	12:63781990-	Deletion	1
36	12:69275207+	12:59253233+	Inversion	1
37	12:69097149+	12:69096468-	Deletion	1
38	12:69098994+	12:69177530-	Deletion	1
39	12:71235024-	12:69253844+	Circularization	1
40	12:76138232-	12:69188137-	Inversion	1
41	12:69088346-	12:58401037+	Circularization	1
42	12:69093263+	12:58405079-	Deletion	1
43	12:69097925-	12:58168108+	Circularization	1
44	12:70283256+	12:69189030+	Inversion	1
45	12:70782871-	12:69055976-	Inversion	1
46	12:71124154-	12:69187885-	Deletion	1
47	12:702740-	12:69087283+	Circularization	1
48	12:76358409+	12:69043441-	Deletion	1
49	12:90852026+	12:69070364-	Deletion	1
50	12:92835247-	12:69233984+	Inversion	1
51	12:126697389-	12:69142306-	Inversion	1
52	12:69058312-	12:61699487+	Circularization	1

Graphing Structural Variants

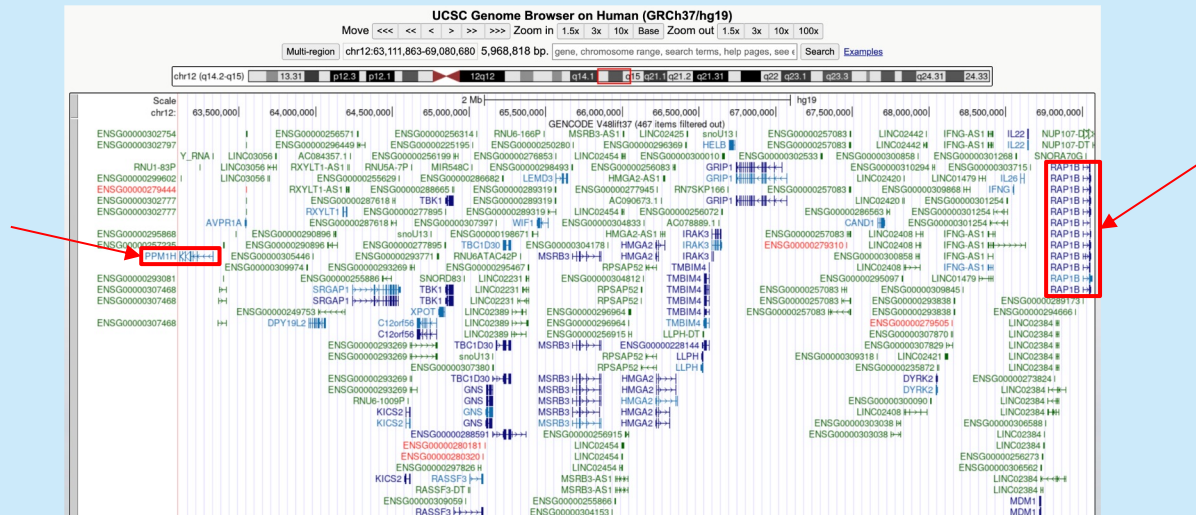
- Using the "filtered_edges.tsv" file from before, we also used "plot_graph.py" to plot arcs representing the discordant edges related to MDM2
- The height of the arc corresponds to the frequency of that edge occurring, and the color designates the type of edge:
 - ++: Blue
 - +/-: Red
 - -+: Brown
 - --: Teal

Graphing Structural Variants



Biological Relevance

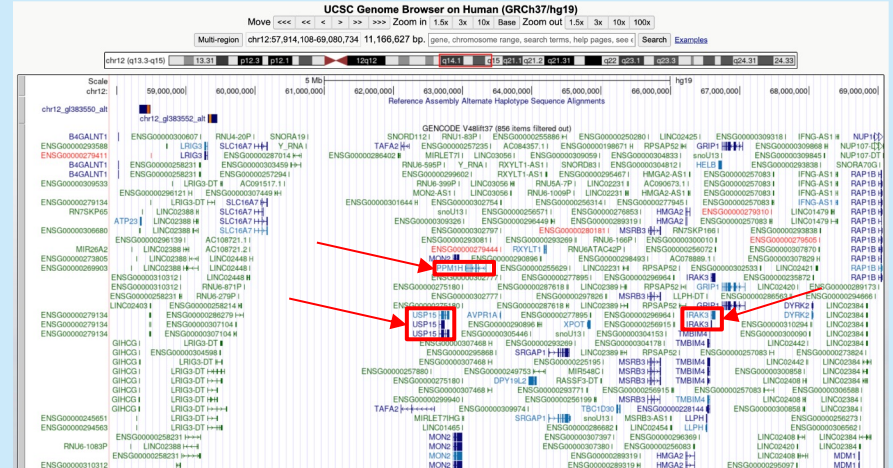
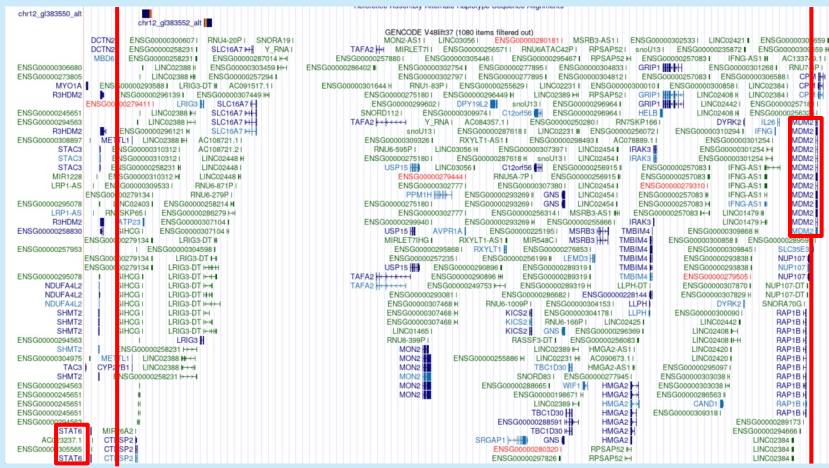
- Most common SV– 12:69080680- → 12:63111863- (Inversion)
 - Occurs 3 times
 - Start position is within the *PPM1H* gene (Protein phosphatase, Mg²⁺/Mn²⁺ dependent 1H)
 - End position is within the *RAP1B* gene (Ras-related protein Rap-1b)



Biological Relevance

Second most common SV: 12:69080734+ → 12:57914108- (Deletion)

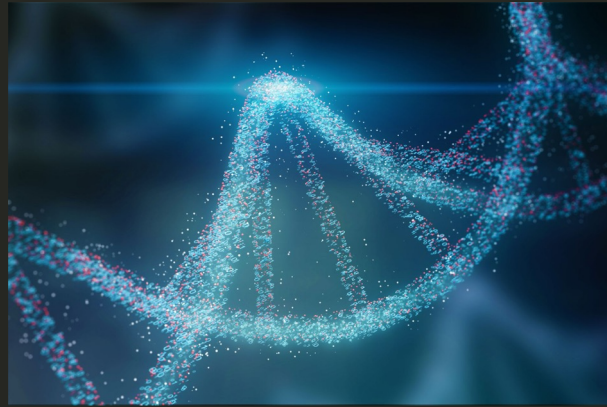
- Occurs 2 times
- 119 genes in the region deleted, i.e. *PPM1H*, *USP15*, *RASSF3*, *IPAK3*
- STAT6* upstream of deleted region, *MDM2* downstream of the region → Possible interaction between the two in ecDNA?



Biological Relevance

- ***PPM1H***: Acts as a tumor suppressor by dephosphorylating proteins involved in cell proliferation and metastasis.
 - ***RAP1B***: Small GTPase that affects tumor growth through regulation of cell proliferation.
 - ***USP15***: Can act as a tumor suppressor by inhibiting Wnt signaling pathway.
 - ***RASSF3***: Ubiquitinates and degrades *MDM2* and increases *p53* levels.
 - ***IRAK3***: Found to have tumor suppressor role in prostate cancer cells.
-
- The loss/change in expression for these genes, particularly with *PPM1H*, in conjunction with *MDM2* expression in ecDNA may promote cancer progression.
 - *STAT6* may induce *MDM2* expression and further promote tumor progression.

Thank you!



-Lydia, Leena, James, Evelyn, & Tim